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You don't have to logically simplify your answers to questions 1 to 4.

1. Let $S \equiv x := x + 1; y := y \% x$ and $q \equiv \text{sqrt}(y) > x$.
 - a. Calculate $wlp(S, q)$.
 - b. Calculate $D(wlp(S, q))$.
 - c. Calculate $D(S)$.
 - d. Calculate $wp(S, q)$.
2. Let $S \equiv \text{if } y \geq 0 \rightarrow x := y/x \ \square \ x \geq 0 \rightarrow x := x/y \text{ fi}$ and $q \equiv x > y \wedge y > z$.
 - a. Calculate $wlp(S, q)$.
 - b. Calculate $D(wlp(S, q))$.
 - c. Calculate $D(S)$.
 - d. Calculate $wp(S, q)$.
3. Calculate $sp(x = y, \text{if } x \geq 0 \rightarrow x := y + 1; z := x \ \square \ x \leq 0 \rightarrow y := x - 1; z := y \text{ fi})$.
4. Calculate $sp(y = x + 1, y := y + 1; \text{if } x < 0 \text{ then } y := -y \text{ fi})$.
5. Let S be a statement and let $q \equiv x > 1$. Decide true or false for the following statements about weakest preconditions ($wlp(S, q)$ or $wp(S, q)$). Justify your answers briefly.
 - a. $\models_{tot} \{wp(S, q)\} S \{q\}$
 - b. If $\sigma \models wlp(S, q)$, then $\perp \notin M(S, \sigma)$.
 - c. There exists some state σ such that $\sigma \not\models wp(S, q)$ but $M(S, \sigma) \models q$.
 - d. Let σ be a well-formed state. If $\sigma \not\models wlp(S, q)$, then $M(S, \sigma) \neq \{\perp\}$.
 - e. Let σ be a well-formed state. If $\sigma \not\models wp(S, q)$, then $\sigma \models \{\neg wp(S, q)\} S \{\neg q\}$.
6. Let $p \equiv x > 1$ and let S be a statement. Decide true or false for the following statements about the strongest postcondition ($sp(p, S)$). Justify your answers briefly.
 - a. $\models_{tot} \{p\} S \{sp(p, S)\}$
 - b. If $M(S, \sigma) - \perp \models sp(p, S)$, then $\sigma \models p$.
 - c. There exists some state σ such that $\sigma \models p$ but $M(S, \sigma) - \perp \not\models sp(p, S)$.
 - d. Let σ be a well-formed state. If $\sigma \not\models p$, then $\sigma \models \{\neg p\} S \{sp(\neg p, S)\}$.
 - e. Let σ be a well-formed state. If $\sigma \not\models p$, then $\sigma \models \{\neg p\} S \{\neg sp(p, S)\}$.
7. Let predicate $p \equiv w * x \neq 0 \wedge z \leq 2 \rightarrow f(w) > 0 \wedge \forall x. \exists y. 0 \leq y \leq x \wedge f(w \div x) + y > f(z)$. Here, w, x, y and z are integer variables. Finish the following syntactic substitutions and show your work.
 - a. $p[y + z / x]$
 - b. $p[x + z / w]$
 - c. $p[x + y / z]$

8. Let x and y be two different integer variables. A student gave the following conjecture:

$$(x * y)[e / x][e' / y] \equiv (x * y)[e' / y][e / x]$$

- Show an example (with some e and some e') in which the above conjecture works.
- Disprove the above conjecture with a counterexample (with some e and some e').